

Inverse & Composite Functions



Composing functions

- 1 Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.
 - 2 Given $f(x) = \sqrt{x + 4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(2)$ and state the domain restriction that makes $f \circ g$ defined.
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Decomposing a composite function

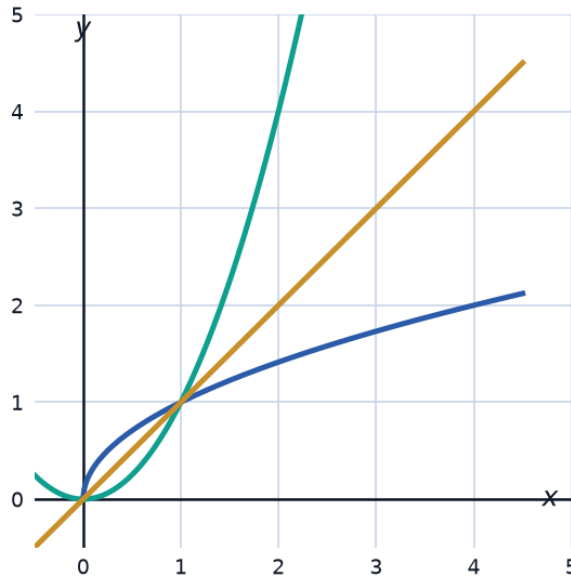
- 3 Write $h(x) = (3x - 5)^4$ as a composition $h(x) = f(g(x))$ for two simpler functions f and g .
 - 4 Write $h(x) = \sqrt{x^2 + 1}$ as a composition of two functions.
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Verifying inverse functions

- 5 Verify algebraically that $f(x) = \frac{x - 2}{3}$ and $g(x) = 3x + 2$ are inverses of each other.
 - 6 Find the inverse of $f(x) = 2x^3 + 1$, and verify that $f(f^{-1}(x)) = x$ for one value, say $x = 17$.
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Visualizing inverse functions as reflections

- 7 The graph shows $f(x) = x^2$ (restricted to $x \geq 0$) and its inverse $f^{-1}(x) = \sqrt{x}$, which are reflections of each other across the line $y = x$.



- a State the domain restriction placed on $f(x) = x^2$ needed for it to have an inverse, and explain why.
- b Confirm from the graph that f and f^{-1} intersect on the line $y = x$, and find that intersection point algebraically.

Domain and range of composite and inverse functions

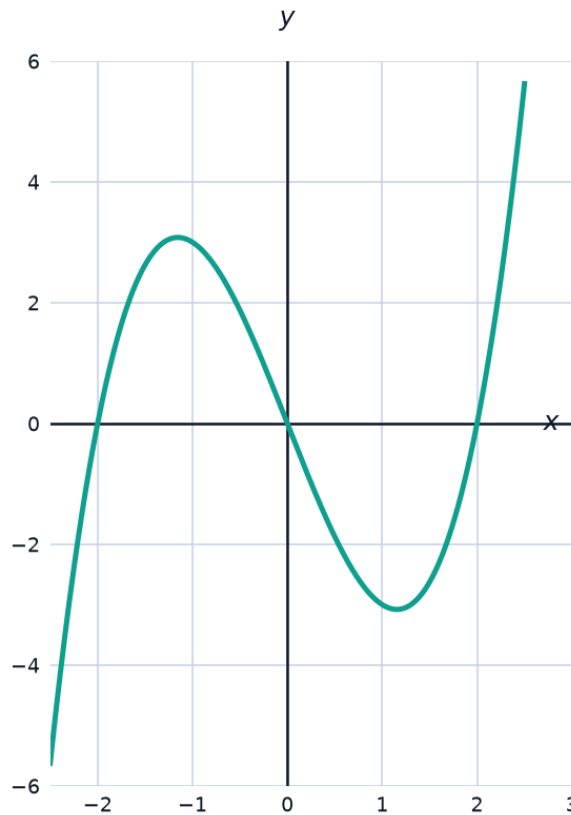
- 8 Given $f(x) = \frac{1}{x-3}$ and $g(x) = x + 1$, find the domain of $(f \circ g)(x)$.
- 9 If f has domain $[1, \infty)$ and range $[0, \infty)$, state the domain and range of f^{-1} .

Applying composite and inverse functions in context

- 10 A temperature-conversion function $C(F) = \frac{5}{9}(F - 32)$ converts Fahrenheit to Celsius. Find $C^{-1}(F)$ and interpret what it computes.

Using the horizontal line test

- 11 Determine whether $f(x) = x^3 - 4x$ has an inverse function over its entire domain. Justify using the horizontal line test.



- 12 Explain why $f(x) = 5x - 2$ passes the horizontal line test for all real x , and find its inverse.

Composite functions with restricted domains

- 13 Given $f(x) = \frac{1}{x}$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and state its domain.

Multiple choice — easy 15

- 14 Given $f(x) = x - 1$ and $g(x) = 5x - 4$, find $(f \circ g)(-6)$.

- a. -34
- b. -36
- c. -35
- d. -39

15 Given $f(x) = -3x - 6$ and $g(x) = 6x$, find $(f \circ g)(-4)$.

a. 67

b. 65

c. 36

d. 66

16 Given $f(x) = 5x - 2$ and $g(x) = x + 2$, find $(f \circ g)(-2)$.

a. -3

b. -10

c. -2

d. -1

17 Given $f(x) = 9x - 3$ and $g(x) = 8x + 1$, find $(f \circ g)(2)$.

a. 151

b. 149

c. 121

d. 150

18 Given $f(x) = 9x + 5$ and $g(x) = -8x - 9$, find $(f \circ g)(-2)$.

a. 69

b. 67

c. 68

d. 95

19 Given $f(x) = 9x - 5$ and $g(x) = -9x - 6$, find $(f \circ g)(-1)$.

- a. 21
- b. 22
- c. 23
- d. 120

20 Given $f(x) = -9x + 2$ and $g(x) = -8x - 1$, find $(f \circ g)(6)$.

- a. 442
- b. 444
- c. 415
- d. 443

21 Given $f(x) = -8x - 6$ and $g(x) = 6x - 7$, find $(f \circ g)(6)$.

- a. -239
- b. -331
- c. -238
- d. -237

22 Given $f(x) = 4x - 1$ and $g(x) = -7x - 3$, find $(f \circ g)(0)$.

- a. -13
- b. -14
- c. 4
- d. -12

23 Given $f(x) = -3x - 2$ and $g(x) = -5x + 1$, find $(f \circ g)(3)$.

a. 56

b. 39

c. 40

d. 41

24 Given $f(x) = 6x - 9$ and $g(x) = -x - 1$, find $(f \circ g)(5)$.

a. -46

b. -22

c. -45

d. -44

25 Given $f(x) = 5x - 6$ and $g(x) = x + 8$, find $(f \circ g)(1)$.

a. 7

b. 39

c. 40

d. 38

26 Given $f(x) = 6x + 4$ and $g(x) = 9x - 6$, find $(f \circ g)(1)$.

a. 84

b. 21

c. 23

d. 22

27 Given $f(x) = -5x - 4$ and $g(x) = 6x - 1$, find $(f \circ g)(6)$.

- a. -179
- b. -178
- c. -205
- d. -180

28 Given $f(x) = -3x - 8$ and $g(x) = 4x - 5$, find $(f \circ g)(1)$.

- a. -4
- b. -6
- c. -5
- d. -49

Multiple choice — medium 15

29 Given $f(x) = -2x - 9$, find $f^{-1}(-5)$.

- a. -2
- b. -3
- c. -1
- d. -5

30 Given $f(x) = 9x - 1$, find $f^{-1}(-73)$.

- a. -8
- b. -73
- c. -7
- d. -9

31 Given $f(x) = 6x - 9$, find $f^{-1}(39)$.

- a. 8
- b. 9
- c. 7
- d. 39

32 Given $f(x) = -4x + 5$, find $f^{-1}(29)$.

- a. -6
- b. -7
- c. -5
- d. 29

33 Given $f(x) = -5x + 4$, find $f^{-1}(44)$.

- a. 44
- b. -7
- c. -8
- d. -9

34 Given $f(x) = 2x + 6$, find $f^{-1}(-4)$.

- a. -6
- b. -5
- c. -4
- d. -7

35 Given $f(x) = -3x + 8$, find $f^{-1}(8)$.

a. 8

b. 0

c. -1

d. 1

36 Given $f(x) = -8x + 7$, find $f^{-1}(-25)$.

a. 4

b. 3

c. -25

d. 5

37 Given $f(x) = -9x - 5$, find $f^{-1}(-68)$.

a. 8

b. 7

c. 6

d. -68

38 Given $f(x) = 6x + 7$, find $f^{-1}(-11)$.

a. -11

b. -3

c. -4

d. -2

39 Given $f(x) = 7x + 7$, find $f^{-1}(-21)$.

a. -21

b. -3

c. -5

d. -4

40 Given $f(x) = -5x + 8$, find $f^{-1}(23)$.

a. -3

b. -2

c. 23

d. -4

41 Given $f(x) = 2x - 9$, find $f^{-1}(-19)$.

a. -5

b. -4

c. -19

d. -6

42 Given $f(x) = 8x - 3$, find $f^{-1}(-3)$.

a. -1

b. 1

c. -3

d. 0

43 Given $f(x) = -9x + 4$, find $f^{-1}(4)$.

- a. -1
- b. 1
- c. 4
- d. 0

Multiple choice — hard 15

44 Given $f(x) = 5x - 7$ and $g(x) = -x^2 + 5$, find $(g \circ f)(-3)$.

- a. -477
- b. 27
- c. -481
- d. -479

45 Given $f(x) = -5x - 8$ and $g(x) = 5x^2 - 6$, find $(g \circ f)(-1)$.

- a. 41
- b. -21
- c. 37
- d. 39

46 Given $f(x) = -4x + 4$ and $g(x) = -x^2 + 7$, find $(g \circ f)(-3)$.

- a. -247
- b. -249
- c. -9
- d. -251

- 47 Given $f(x) = -2x - 1$ and $g(x) = 5x^2$, find $(g \circ f)(1)$.
- a. 45
 - b. 43
 - c. -15
 - d. 47
- 48 Given $f(x) = -3x - 8$ and $g(x) = 3x^2 - 4$, find $(g \circ f)(0)$.
- a. 190
 - b. 188
 - c. -28
 - d. 186
- 49 Given $f(x) = -7x$, find the slope and y -intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).
- a. $(-7, 0)$
 - b. $(-\frac{1}{7}, 0)$
 - c. $(\frac{1}{7}, 0)$
 - d. $(\frac{5}{14}, 0)$
- 50 Given $f(x) = 5x - 5$, find the slope and y -intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).
- a. $(\frac{1}{5}, -1)$
 - b. $(\frac{1}{5}, 1)$
 - c. $(5, -5)$
 - d. $(-\frac{1}{5}, 1)$

- 51 Given $f(x) = -9x - 1$, find the slope and y -intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).
- a. $(\frac{1}{9}, -\frac{1}{9})$
 - b. $(-9, -1)$
 - c. $(-\frac{1}{9}, -\frac{1}{9})$
 - d. $(-\frac{1}{9}, \frac{1}{9})$
- 52 Given $f(x) = 2x + 4$, find the slope and y -intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).
- a. $(\frac{1}{2}, -2)$
 - b. $(-\frac{1}{2}, -2)$
 - c. $(2, 4)$
 - d. $(\frac{1}{2}, 2)$
- 53 Given $f(x) = 6x - 5$, find the slope and y -intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).
- a. $(-\frac{1}{6}, \frac{5}{6})$
 - b. $(6, -5)$
 - c. $(\frac{1}{6}, -\frac{5}{6})$
 - d. $(\frac{1}{6}, \frac{5}{6})$
- 54 Determine whether $f(x) = x^3 - 6x$ has an inverse function over its entire domain, using the horizontal line test.
- a. Has an inverse function over its entire domain (strictly increasing)
 - b. Does NOT have an inverse function over its entire domain (not one-to-one)
 - c. Has an inverse only on the interval $x \geq 0$
 - d. Cannot be determined without a graph

- 55 Determine whether $f(x) = x^3 - 3x$ has an inverse function over its entire domain, using the horizontal line test.
- a. Does NOT have an inverse function over its entire domain (not one-to-one)
 - b. Has an inverse only on the interval $x \geq 0$
 - c. Cannot be determined without a graph
 - d. Has an inverse function over its entire domain (strictly increasing)
- 56 Determine whether $f(x) = x^3 + x$ has an inverse function over its entire domain, using the horizontal line test.
- a. Has an inverse only on the interval $x \geq 0$
 - b. Has an inverse function over its entire domain (strictly increasing)
 - c. Does NOT have an inverse function over its entire domain (not one-to-one)
 - d. Cannot be determined without a graph
- 57 Determine whether $f(x) = x^3 - 2x$ has an inverse function over its entire domain, using the horizontal line test.
- a. Does NOT have an inverse function over its entire domain (not one-to-one)
 - b. Cannot be determined without a graph
 - c. Has an inverse function over its entire domain (strictly increasing)
 - d. Has an inverse only on the interval $x \geq 0$
- 58 Determine whether $f(x) = x^3 + 2x$ has an inverse function over its entire domain, using the horizontal line test.
- a. Has an inverse only on the interval $x \geq 0$
 - b. Does NOT have an inverse function over its entire domain (not one-to-one)
 - c. Cannot be determined without a graph
 - d. Has an inverse function over its entire domain (strictly increasing)